Truth-Tracking Profiles: What LLMs Participate In

Brett Reynolds [*]
Humber Polytechnic & University of Toronto

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Abstract

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1 Introduction: a binary presupposition

Do the representations inside a large language model connect to the world? The question is usually put as a yes/no. Bender and Koller (2020) say no: a system trained on form alone, like an octopus eavesdropping on telegraph traffic, learns the statistics of language without touching what language is about. Others reply that such systems meet at least some conditions on reference, or that sensory grounding isn't required for thought in the first place (Chalmers, 2023; Coelho Mollo & Millière, 2026). The camps disagree on the verdict but share the question's shape.

That yes/no framing makes world-connection look like a single capacity, something a system either has or lacks. The successes we call truth-tracking are instead produced by distinct stabilising mechanisms that ordinarily operate together: perceptual contact, testimonial inheritance, inferential coherence, measurement and intervention, error-correction, and practical success. A system can take part in some of these while sitting out the rest, which is just what a language model does.

The paper's claim is that those stabilisers form a projectible profile around the representational successes picked out by the predicate *true*: a pattern whose point is prediction and explanation rather than classification for its own sake (Goodman, 1955; Khalidi, 2013). LLMs matter because they make the dissociation visible: they inherit dense testimonial traces and local coherence patterns while lacking much of the perceptual and verificational coupling that normally keeps claims answerable to the world.

The contribution is diagnostic. Boyd (2021) treats reference and truth as special cases of accommodation: the alignment of perceptual, instrumental, cognitive, and representational practices with causally relevant structures. This paper asks what happens when a system enters some of that alignment through human text while missing other routes through which accommodation is normally maintained. LLMs inherit textual products of testimony and coherence, while lacking or weakening

[*]Contact: brett.reynolds@humber.ca

perception, measurement, intervention, and world-directed correction. The profile view predicts that this partial participation should have a characteristic failure signature.

That diagnostic has to be earned, because the profile view invites an obvious complaint. The component theories each isolate a genuine stabiliser. Reliabilism gets process reliability right (Goldman, 1979); coherentism gets inferential integration and the demand for truth-conducive input right (BonJour, 1985); pragmatism gets the corrective role of action and success right; social epistemology gets testimonial and institutional transmission right (Coady, 1994; Goldberg, 2010; Lackey, 2008). The relabeling worry is that profile talk merely bundles these local insights and calls the bundle "projectible". If that's all the view does, it adds nothing.

It does more, in three ways no single-mechanism theory can match. First and most sharply, the profile structure predicts a characteristic failure signature: pull the stabilisers apart, and the account says in advance what should go wrong. Second, it makes conflicts between mechanisms adjudicable, by treating their degree of co-variation as evidence in its own right. Third, it predicts where truth-relevant mechanisms should co-vary tightly across domains and where they should come apart.

The language-model case is where the first prediction pays off. A text-only system trained on human writing is strongly exposed to the textual traces of testimony and to local patterns of coherence, but it is not thereby a normal speaker or hearer in a testimonial exchange, and it lacks direct perceptual contact, independent measurement, and ordinary correction through action in the world. The profile view therefore predicts a distinctive vulnerability: fluent and internally coherent outputs can track textual regularities while failing to remain answerable to the facts those regularities were originally used to report.

Human beings hallucinate, misremember, and confabulate too. The contrast lies in the profile of correction. In ordinary human cases, those failures occur against a background of perception, action, and social correction; in a text-only model, several of those world-corrective paths are absent or attenuated as part of the standing profile. What the LLM literature calls hallucination is one expression of that vulnerability (Huang et al., 2025).

That vulnerability marks the distance between the formal linguistic competence these models plainly have and the functional competence that world-contact underwrites (Mahowald et al., 2024).

The argument runs as follows. Section 2 fixes the target: the relevant cases are representational successes (beliefs, utterances, measurements, models, inferential outputs), not truth as an abstract object. Section 3 sets out the stabilising mechanisms and their Boydian lineage.

Section 4 says why the profile projects, locating projectibility in goal achievement under counterfactual perturbation rather than in survival alone. Section 5 makes good on the adjudication and co-variation claims, the core of the reply to relabeling. Section 6 takes objections. Section 7 develops the language-model profile mechanism by mechanism. Section 8 concludes.

2 The target: what *true* picks out

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3 Truth-tracking as a projectible mechanism profile

Goodman's new riddle fixes the projectibility problem: the hard question isn't whether past cases support some generalisation, but which predicates support projection from known to unknown cases

(Goodman, 1955). The present paper asks that question about truth-tracking stabilisers rather than about one predicate in isolation.

Khalidi's account supplies the network discipline behind this profile talk. Natural categories earn their status by supporting prediction and explanation, and they do so when associated properties stand in causal relations that make some properties projectible from others (Khalidi, 2013).

The present use is an adaptation: rather than treating the stabilisers of truth-tracking as natural kinds in their own right, it represents their relations as a causal profile. Perception, testimony, measurement, coherence, intervention, correction, and practical success are linked routes through which representational outputs become more or less answerable to the world. Khalidi's later "nodes in causal networks" formulation gives the graph vocabulary for this representation (Khalidi, 2018).

This profile view is Boydian in lineage. Homeostatic property cluster (HPC) theory explains how real kinds can be projectible without essences (Boyd, 1991, 1999). Boyd's later accommodationism supplies the semantic background: successful reference is an achievement of whole practices, not a relation fixed term by term by accepted definitions. Perceptual skills, instruments, inferential routines, social arrangements, and representational uses all contribute when they help a practice align with relevant causal structures (Boyd, 2021).

The present claim is narrower than Boyd's. It asks how the accommodation profile changes when the participant is a text-trained system. An LLM can inherit stabilized linguistic and testimonial patterns while missing the instrumental, perceptual, and practical routes that normally keep those patterns answerable. Stable or projectible profiles shouldn't automatically be called homeostatic; that stronger description should be reserved for cases where corrective coupling has been shown (Reynolds, 2026).

That means the paper is continuous with accommodationism at the explanatory level and diagnostic at the level of system profile. Accommodationism tells us why representational practices can be answerable to causal structure; the profile account tracks how strongly a given system participates in the stabilisers that make that answerability possible. LLMs are the pressure case because they inherit accommodation through human text while missing or weakening several world-corrective paths.

Boyd's homeostatic property cluster is the important special case in which links are actively maintained, where mechanisms correct one another so that the properties co-occur far more reliably than chance. The truth-tracking profile has tightly coupled regions, and the connections thin toward its edges. The profile view doesn't need the whole field to be literally homeostatic before it can project.

That picture does two jobs. The tightly coupled core, everyday perception and basic coordination, is where mutual correction is strongest, and the coupling there is what predicts the language model's failure signature: remove the perceptual and verificational nodes and the profile comes apart in a specific way (Section 7).

It places the loose cases. Toward the edges, in theoretical physics, ethics, and large-number mathematics, the nodes are real but sparsely linked. Reading the profile as a network doesn't by itself fix where its boundaries fall: Onishi and Serpico (2022) show that carving a kind from a causal graph stays relative to the chosen level of abstraction, the interest-relativity Craver (2009) found in mechanism talk.

So the paper fixes the epistemic purpose. Following Khalidi's distinction between epistemic purposes and merely practical or aesthetic ones, the profile is individuated by representing the shared world well enough for goal achievement under perturbation (Khalidi, 2013). Relative to that purpose, its connections can thin at the edges without the profile dissolving.

4 Why the profile is projectible

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5 What the profile structure does that component theories don't

This section makes good on the claim from Section 1 that the profile view earns its keep. The test isn't whether "truth depends on many mechanisms" sounds plausible; it's whether the profile structure answers questions that reliabilism, coherentism, pragmatism, and correspondence can't pose from inside their own resources.

The component theories enter here as local successes: each marks one stabiliser. The profile claim earns its keep only where couplings among those stabilisers support predictions a single-component theory cannot formulate.

Read as a causal network (Section 3), the profile has structure a single-mechanism theory cannot represent: the strength of the links between its nodes. Two questions turn on that structure; a third, the language-model failure signature, is taken up in Section 7.

5.1 Adjudicating mechanism conflict

When two stabilisers disagree, say perception reports one thing and well-attested testimony another, a theory focused on one source at a time underdescribes the evidential situation. Goldman-style reliabilism can ask which process type is reliable, including conditional reliability; social reliabilism can extend the assessed process through testimonial production (Goldberg, 2010; Goldman, 1979). Those are genuine resources. The profile question is different: how tightly have these sources been calibrated against one another in this domain? The network view adds a resource the component theories lack: the strength of the links between nodes.

In Khalidi's terms, causal relations among associated properties underwrite projection; here the relevant projection runs from one stabiliser's output to another's (Khalidi, 2013).

Where the stabilisers are densely connected and have tracked one another, a lone dissenter is probably in error; where the links are weak, no single output earns much trust.

5.2 Domain-clustered co-variation

The network view also predicts a cross-domain pattern no one-mechanism theory can generate: a theory that privileges one node has little to say about how connectivity varies from one region to another. Khalidi treats causal strictness and network density as dimensions along which categories support stronger or weaker projection (Khalidi, 2013). The stabilisers should be densely linked where they've been jointly disciplined (everyday perception, basic coordination) and sparsely linked where they haven't (theoretical physics, ethics, large-number mathematics). Persistent disagreement should cluster in the sparsely connected regions.

The prediction is non-circular only if connectivity is characterised without appeal to whether the mechanisms agree, and it can't be read off the graph in a purpose-free way: Onishi and Serpico (2022) show the abstraction level and boundaries stay interest-relative.

The reply is to fix the purpose rather than disown it. Holding the representational aim of Section 4 fixed, connectivity is characterised by shared selection history, institutional age, and measurement infrastructure, each measurable without checking whether the mechanisms agree, and those measures are what predict where disagreement clusters.

6 Objections

6.1 Interest-relativity

One objection cuts at the profile structure itself. Onishi and Serpico (2022) argue that Khalidi's causal-network account inherits the interest-relativity Craver (2009) finds in mechanistic accounts: which causal graph you draw, at what level of abstraction, and where you set its boundaries all answer to explanatory interest, so the network never fixes a unique, interest-free carving of kinds. If that's right, truth-tracking-as-a-profile is only as objective as our purposes.

The reply concedes the premise and denies the damage. The account never needed an interest-free carving. It makes two claims, both purpose-relative and both objective: a system missing the perceptual and verificational stabilisers shows the failure signature of Section 7, and connectivity tracks selection history, institutional age, and measurement infrastructure (Section 5). Each holds once the representational purpose is fixed, and neither requires that the purpose be read off nature.

Onishi and Serpico (2022) reach the same general verdict, that interest-relativity is ineliminable, and Khalidi's domain pluralism leaves room for this move (Khalidi, 2013). This account makes the operative interest explicit instead of disowning it.

7 The LLM payoff: participation, not a verdict

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8 Conclusion

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